

MA341

APPLIED ODE'S

Homework 4.1

1. In the following questions you are given a homogeneous ODE together with one of its solutions y_1 . Use Wronskian and Abel's formula to find y_2 .

- (a) $xy'' - (x+1)y' + y = 0$, $y_1 = e^x$
- (b) $x^2y'' - 4xy' + 6y = 0$, $y_1 = x^2$
- (c) $xy'' + (5x-1)y' - 5y = 0$, $y_1 = 5x-1$
- (d) $xy'' + (1-2x)y' + (x-1)y = 0$, $y_1 = e^x$

2. Find a general solution for the following ODEs:

- (a) $6y'' - y' - y = 0$
- (b) $y'' - 2y' - 2y = 0$
- (c) $y'' - 2y' + 6y = 0$
- (d) $y'' + 5y' = 0$
- (e) $y'' - 8y' + 16y = 0$

3. Consider the following initial value problem

$$\begin{cases} y'' - y' - 2y &= 0 \\ y(0) &= \alpha \\ y'(0) &= 2 \end{cases}$$

- (a) Find solution to the problem.
- (b) What values of α gives $\lim_{x \rightarrow \infty} y(x) = 0$.

4. Use D'Alembert Order Reduction Method to find solution to the following ODEs.

- (a) Find y_1 by yourself, then reduce the order and solve the resulting ODE to find y_2 , the general and finally the private solution.

$$\begin{cases} 9y'' + 6y' + y &= 0 \\ y(0) &= 3 \\ y'(0) &= 0 \end{cases}.$$

- (b) You are going to solve the following third order ODE

$$\begin{cases} y''' - 3y'' + 3y' - y &= 0 \\ y(0) &= 1 \\ y'(0) &= 1 \\ y''(0) &= 1 \end{cases}.$$

First, use a known private solution $y_1 = e^x$ (verify it is indeed a solution!) to reduce its order, then solve resulting 2nd order ODE. Use it to find the solution of the original problem.